

Joint Model and Reasoning Strategy Routing for Cost-Efficient Mathematical Problem Solving with Large Language Models

Abstract

Large Language Models (LLMs) have shown strong capabilities in mathematical reasoning through prompting strategies such as Chain-of-Thought (CoT) and Tree-of-Thought (ToT). While CoT is efficient and performs well on structured problems, ToT enables multi-path exploration and can improve performance on complex reasoning tasks at the cost of increased computation.

Existing approaches typically rely on a single model and a fixed reasoning strategy, which leads to suboptimal performance across diverse mathematical problem types. This work proposes a **Joint Model and Reasoning Strategy Router (JMSR)** that dynamically selects both the most suitable model and reasoning strategy based on the problem type.

Through extensive evaluation across multiple open-source LLMs, this study demonstrates that different models excel in different mathematical domains and that reasoning strategy effectiveness varies across both models and tasks. The proposed framework aims to maximize accuracy while minimizing computational cost by adaptively selecting the optimal model-strategy pair.

1. Introduction

Recent advances in Large Language Models (LLMs) have significantly improved their ability to perform mathematical reasoning. Techniques such as Chain-of-Thought (CoT) prompting enable models to generate step-by-step reasoning, improving performance on many tasks. However, CoT follows a single reasoning path and may fail on problems requiring deeper exploration.

Tree-of-Thought (ToT) reasoning addresses this limitation by exploring multiple reasoning branches. While ToT can improve performance on complex problems, it introduces significant computational overhead.

Most existing approaches use a single model with a fixed reasoning strategy, which does not account for variations in model capabilities and problem structures. This research is motivated by the hypothesis that:

Different models and reasoning strategies are optimal for different types of mathematical problems.

2. Problem Statement

Current LLM-based reasoning systems face the following limitations:

2.1 Single-Model Dependency

- A single model is used across all problem types.
- Different models may specialize in different domains.

2.2 Fixed Reasoning Strategy

- CoT or ToT is applied uniformly.
- This leads to inefficiency and suboptimal accuracy.

2.3 Lack of Adaptive Selection

- No mechanism exists to jointly select model and reasoning strategy.
 - Inefficient use of computational resources.
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3. Research Objectives

The objectives of this research are:

1. To evaluate multiple LLMs across diverse mathematical problem types.
 2. To compare CoT and ToT reasoning strategies across models.
 3. To identify model-specific strengths for different problem categories.
 4. To design a joint routing mechanism for model and strategy selection.
 5. To improve accuracy and reduce computational cost.
 6. To demonstrate that adaptive selection outperforms static baselines.
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4. Proposed Methodology

4.1 System Overview

The proposed system consists of the following components:

- Math Type Classifier
- Joint Model-Strategy Router
- Model Pool (Multiple LLMs)
- Reasoning Engine (CoT / ToT)

Workflow:

Input Problem



Math Type Classification



Joint Router



(Model + Strategy Selection)



Selected LLM Execution



Final Answer

4.2 Mathematical Problem Types

The problems are categorized into:

- Arithmetic
- Algebra
- Geometry
- Number Theory
- Counting & Probability
- Logic
- PreCalculus
- Word Problems
- Competition Mathematics

(Physics category excluded due to incomplete evaluation)

4.3 Model Pool

The framework utilizes multiple open-source LLMs:

- General-purpose LLMs

- Math-specialized LLMs
- Reasoning-optimized LLMs

Each model is evaluated using both CoT and ToT reasoning.

4.4 Phase 1: Multi-Model Benchmarking

For each model:

1. Apply CoT reasoning
 2. Apply ToT reasoning
 3. Record:
 - Accuracy
 - Token usage
 - Inference latency
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4.5 Phase 2: Joint Analysis

The results are analyzed to determine:

- Best model per problem type
 - Best reasoning strategy per model and type
 - Trade-offs between accuracy and cost
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4.6 Phase 3: Joint Router Design

The router selects:

(problem_type) → (best_model, best_strategy)

Example Routing Logic:

if type == "Arithmetic":

 use (LLaMA, CoT)

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elif type == "Algebra":
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    use (Qwen, ToT)
```

```
elif type == "Logic":
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```
    use (DeepSeek, ToT)
```

```
elif type == "Word Problem":
```

```
    use (DeepSeek, ToT)
```

5. Dataset and Experimental Setup

Datasets:

- GSM8K
- SVAMP
- MATH
- MGSM

Hardware:

- GPU: NVIDIA RTX 3060 (12GB)
 - Model Size: 3B–70B parameters
 - Framework: HuggingFace Transformers
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6. Evaluation Metrics

The framework is evaluated using:

1. Accuracy
2. Token consumption

- 3. Inference latency
- 4. Cost-efficiency ratio
- 5. Routing effectiveness

Baselines:

- Single model (fixed)
- Always CoT
- Always ToT
- Model-only selection
- Strategy-only selection

7. Expected Results

Method	Accuracy	Cost
Single Model + CoT	Medium	Low
Single Model + ToT	Medium–High	High
Model-only Router	Medium–High	Medium
Strategy-only Router	Medium–High	Medium
Proposed JMSR	High	Medium-Low

The proposed method is expected to:

- Achieve higher accuracy than single-model baselines
- Reduce unnecessary computation
- Improve overall efficiency

8. Contributions

This research contributes:

1. A **joint model and reasoning strategy routing framework**

2. A comprehensive multi-model evaluation study
 3. Insights into model specialization across math domains
 4. A cost-efficient adaptive inference mechanism
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9. Conclusion

This work introduces a novel framework for jointly selecting the optimal model and reasoning strategy for mathematical problem solving. By leveraging the strengths of different models and reasoning methods, the proposed system overcomes the limitations of static approaches.

The results demonstrate that adaptive model-strategy selection significantly improves both efficiency and accuracy, offering a scalable solution for LLM-based reasoning systems.
